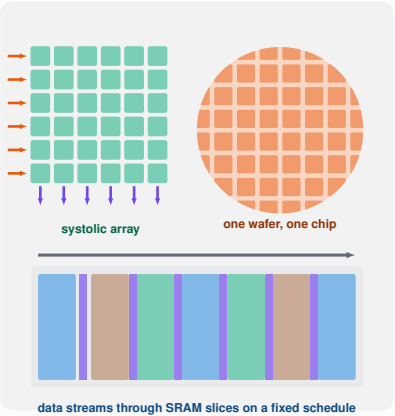
LECTURE NOTES · HARDWARE, 2026

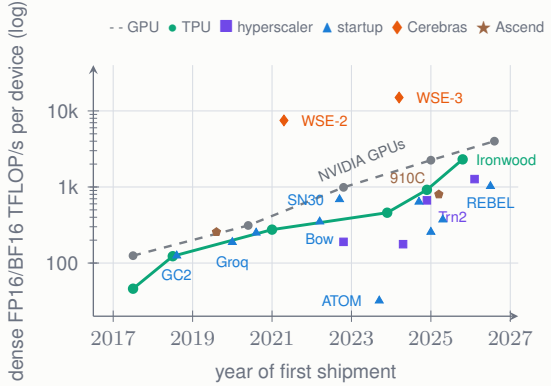
Non-GPU AI Accelerators

Systolic arrays, dataflow engines, wafers and SRAM machines, 2016–2027: Google's TPUs; Cerebras, SambaNova, Groq, Graphcore, Tenstorrent, Rebellions and Furiosa; the hyperscalers' own chips; China's Ascend, Kunlun and Cambricon; what is emerging, what the benchmarks measure, and why most of these designs did not survive.



A decade of accelerators that are not GPUs

Dense 16-bit peak per device by year shipped; the GPU line is the reference every one of them was built to beat.



How to read it[†]

One unit of account: dense 16-bit matrix peak per package; FP8 and FP4 double and quadruple it where supported (Ironwood 4.6 PF FP8). Sparsity multipliers removed.

Two lines track the GPU: the TPU and the hyperscaler ASICs sit a year or two behind NVIDIA per chip and make it up in pods and price. The startups mostly sit below the line and sell something else: latency (Groq, Cerebras), memory tiers (SambaNova), openness (Tenstorrent), locality (Rebellions, Furiosa, Ascend). Each Cerebras point is a whole wafer.

Not on the chart: SRAM bandwidth, where the SRAM-resident designs win (slides 4–8), and software, which decided most fates (slide 35).

Who is who: fifteen programs, 2016–2027

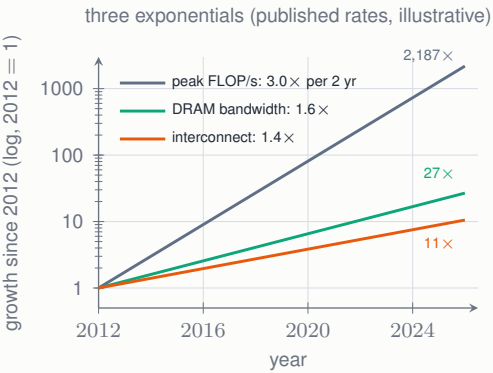
First and latest chips, the design idea, and where each program stood in November 2026.

program[†]	first chip	latest chip	design idea	status, Nov 2026
Google TPU	v1, 2015 (paper 2017)	v7 Ironwood, 2025; TPU 8 ann.	systolic MXUs, optical torus	Gemini; Anthropic $\leq$1 GW; Meta rents
Cerebras	WSE-1 / CS-1, 2019	WSE-3, 2024; CS-4, 2026	one wafer, 44 GB SRAM	IPO May 2026; OpenAI 750 MW
Groq	TSP, 2020	LPU (same silicon); GroqCloud	SRAM-only, static streaming	IP to NVIDIA, $20B, Dec 2025
SambaNova	SN10, 2020	SN40L, 2024; SN50, 2026	dataflow CGRA, 3-tier memory	$1B at $11B, Jul 2026
Graphcore	GC2 IPU, 2018	Bow IPU, 2022	1,472 MIMD tiles, 900 MB SRAM	SoftBank, Jul 2024 ($\sim$$600M)
Tenstorrent	Grayskull, 2021	Blackhole, 2025	Tensix cores, RISC-V, open SW	$693M, Dec 2024; licenses IP
Rebellions	ATOM, 2023	REBEL, 2025–26	inference NPU, chiplets, HBM3e	merged with Sapeon, Dec 2024
FuriosaAI	Warboy, 2021	RNGD, 2025	tensor contraction processor	declined Meta, Mar 2025; LG
AWS	Inferentia, 2019	Trainium3, 2025	systolic NeuronCores, HBM	$\sim$1M Trn2; 500k+ Anthropic
Microsoft	Maia 100, 2024	Maia 200, 2026	MX tiles, Ethernet fabric	Azure inference; OpenAI IP
Meta	MTIA v1, 2023	MTIA v2, 2024; 300 series	RISC-V PEs, SRAM, LPDDR5	ranking; buys TPUs too
Huawei	Ascend 910, 2019	910C, 2025; 950PR, 2026	Da Vinci cube; SuperPoDs	CloudMatrix 384; China's default
Intel Habana	Gaudi, 2019	Gaudi 3, 2024	MME + TPCs, Ethernet on chip	line ended 2025
Qualcomm	Cloud AI 100, 2021	AI200, 2026; AI250, 2027	NPU cores, LPDDR capacity	200 MW with Humain
d-Matrix	Corsair, 2024	Raptor, 2026	digital in-memory compute	$275M at $2B, Nov 2025

first shipments or general availability where known, otherwise announcements; business facts from filings and press (sources on the section slides); omitted: Tesla Dojo (paused 2025), Nervana and Wave (2020), Untether (2025), the photonic and analog companies (slides 28–30), Kunlun, Hanguang and Cambricon (slide 27)

The memory wall: why the GPU's critics had a point

Peak compute grew 3× every two years, DRAM bandwidth 1.6×, interconnect 1.4×; decode is bandwidth-bound.



The argument every non-GPU company made†

Decode is a memory problem: each token reads every weight once; at batch 1 a 70B model in 8 bits moves 70 GB per token, so an H100 (3.35 TB/s) cannot pass about 48 tokens/s whatever its tensor cores do (derived).

The GPU answer: batch users so one weight read serves many tokens; HBM3e, then HBM4; NVLink to pool memory.

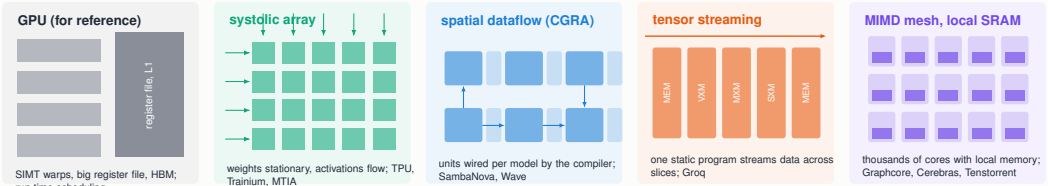
The alternative: keep weights in on-chip SRAM at tens of PB/s and accept that capacity is the constraint (slide 8).

Or change the math unit: systolic or dataflow fabrics pass operands unit to unit, not through a register file, for less energy per multiply-add (slides 5 and 7).

Rates: Gholami et al., "AI and memory wall" (2024).

A taxonomy: five ways to organize a tensor machine

Where the operands wait and who decides when they move: in hardware, in a compiler schedule, or in a program.



How to read the five†

Who schedules: the GPU at run time, warp by warp; the systolic array in hardware; the streaming processor and the CGRA in the compiler; the MIMD mesh in programs that synchronize in bulk-synchronous steps (Valiant, 1990).

Where the weights sit: HBM (GPU, TPU, Trainium); on-chip SRAM (Groq, Graphcore, Cerebras); SRAM, HBM and DDR tiers (SambaNova).

The trade each one makes†

Systolic: best utilization on large dense matrix multiplies, poor on small or ragged ones.

Dataflow, streaming: no caches or reordering, so batch-1 latency is low and predictable; each new operator needs compiler work.

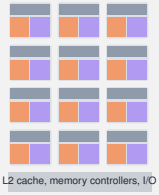
MIMD mesh: flexible, good on irregular work; capacity limited to SRAM. **GPU:** best at nothing on paper, adequate at everything.

Why not a GPU? What a special-purpose chip gives up

TPU v1 (2017): 15–30× a K80 and 30–80× its perf/W, by dropping what a GPU keeps.

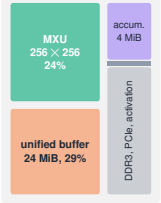
where the silicon goes: three floorplans (illustrative; TPU v1 shares from Jouppi et al.)†

GPU (SIMT)



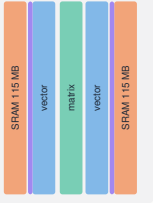
each SM: scheduler, registers, math units; caches at every level

TPU v1 (2015)



control: 2% of the die; no caches, no speculation; 5 main instructions

Groq TSP (2020)



slices span the die; data streams east–west on a compile-time schedule

the GPU spends area on finding parallelism at run time (schedulers, scoreboards, caches, registers for tens of thousands of threads); the ASICs spend it on arithmetic and on-chip memory and let the compiler find the parallelism. Jouppi et al. (ISCA 2017): 92 TOPS INT8, 28 MiB on-chip, 40 W measured, 15–30× the K80 and Haswell on inference, 30–80× per watt†

Given up, and gained

Given up: (1) *dynamic control*: no warp schedulers, speculation or scoreboards; a program is a static schedule the compiler proves correct. (2) *The cache hierarchy*: software-managed buffers whose contents and timing are known in advance. (3) *Generality*: a fixed set of large operators; new ops wait for the compiler.

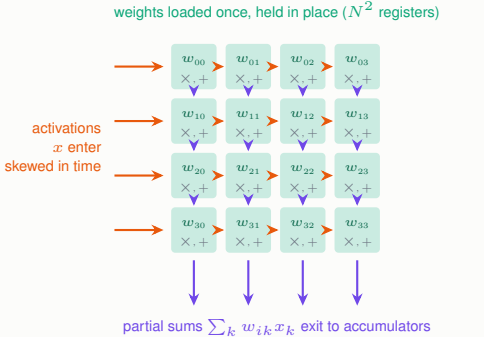
Gained: more MACs and more SRAM per mm^2; one operand fetch feeds hundreds of multiplies; predictable latency, which serving fleets are sized to; better perf/W at a given node.

The bill comes due in software: each design needed a compiler that maps PyTorch graphs onto a static schedule, and most companies on the next slides spent more on that than on silicon. The GPU keeps winning where models change faster than compilers.

Systolic arrays and the MXU: reuse by geometry

16,384 multiply-adds per clock from one operand read per row; the TPU has kept the idea for seven generations.

a weight-stationary systolic array (4×4 shown; the MXU is 128×128 or 256×256)



peak = arrays $\times N^2 \times 2 \times f$: TPU v1: $1 \times 256^2 \times 2 \times 0.7$ GHz = 92 TOPS; TPU v4: 2 TensorCores × 4 MXUs $\times 128^2 \times 2 \times 1.05$ GHz = 275 TFLOP/s BF16 (derived). Operand traffic per MAC falls as $1/N$: the array needs $2N$ inputs per clock for N^2 multiplies.

Why it works, and where it hurts

Reuse by wiring: each weight is used by every activation that passes, each activation by every weight in its row; the register file and crossbar a GPU needs to feed 128 lanes become nearest-neighbor wires.

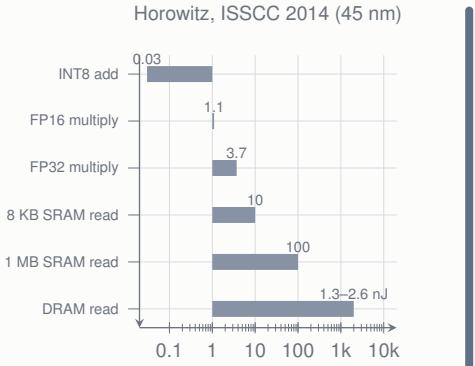
Utilization is the catch: layers with fewer than N outputs, batches smaller than N, or odd shapes leave cells idle; Norrie et al. (Hot Chips 2020) give this as the reason TPU v2 moved from one 256×256 array to several 128×128 arrays.

Loading weights takes N cycles per tile, so tiles are double-buffered; small matrices (decode, short attention) still starve the array, so inference chips add big vector units.

GPUs converged halfway: a tensor core is a small matrix unit under SIMT control, more TPU-like each generation.

SRAM against HBM: bandwidth is cheap on chip, capacity isn't

An on-chip read costs a hundredth of a DRAM access; a 70B model on chip takes hundreds of chips or a wafer.



a DRAM access costs about 500 FP32 multiplies; the numbers are 45 nm and 2014, the ratios have widened since (Dally, CACM 2020)

How many chips hold a 70B model? (derived)†

Weights: 70 GB at 8 bits, 140 GB at 16, plus KV cache.

H100 (80 GB HBM): one GPU at 8 bits, two at 16; 3.35 TB/s each.

Groq LPU (230 MB SRAM, no DRAM): at least 305 chips at 8 bits; Groq served Llama 2 70B from 576 (2024), each reading its slice at 80 TB/s.

Graphcore Bow (900 MB): 78 chips at 8 bits; the IPU-POD streamed weights from host DDR instead.

Cerebras WSE-3 (44 GB): two wafers at 8 bits, four at 16, or one wafer streaming weights from MemoryX.

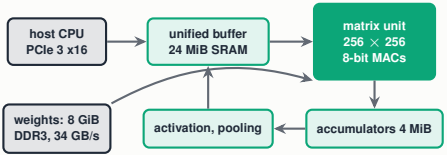
SambaNova SN40L: 520 MB SRAM + 64 GB HBM + 1.5 TB DDR5 per chip; eight hold a trillion parameters.

The rule: SRAM buys bandwidth and latency; capacity comes from chip count, stacking or tiers, each at a cost.

Google TPU v1 to v3: the systolic array goes to the data center

A 2015 inference coprocessor became, by 2018, a liquid-cooled 1,024-chip training pod; the pod is the design unit.

TPUv1 (2015): one chip, one array, no cache†



weights are loaded into the array and stay; activations flow through; 92 TOPS INT8 at 75 W, 15–30× a K80 on 2015 inference (Jouppi et al., ISCA 2017); the roofline showed the 34 GB/s DDR3 as the limit

TPU†	v1	v2	v3
deployed	2015	2017	2018
process	28 nm	16 nm	16 nm
peak per chip	92 TOPS INT8	46 TF BF16	123 TF BF16
memory	8 GiB DDR3	16 GiB HBM	32 GiB HBM
bandwidth	34 GB/s	700 GB/s	900 GB/s
power	75 W	280 W	450 W, liquid
pod	1 chip	256 chips, 11.5 PF	1,024 chips, 100+ PF

What Google learned, and told†

Remove what inference does not need: caches, branch prediction, out-of-order issue; the array and a software-managed buffer gave 30–80× the TOPS per watt of a 2015 CPU or GPU (ISCA 2017).

Deterministic: no caches means a predictable 99th percentile, which is what a serving fleet is sized to.

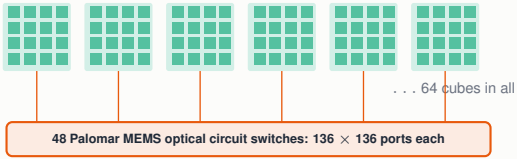
v2 (2017) was a different machine: two TensorCores, each a VLIW scalar unit driving vector units and 128 × 128 matrix units; BF16 invented here; HBM; and an inter-chip interconnect with a router in every chip forming a 2D torus. The pod, not the chip, is designed (Norrie et al., 2021).

v3 (2018): twice the matrix units per core, 450 W, liquid cooling, 1,024 chips. **Lesson:** a fixed array needs a compiler that pads and tiles for it: XLA, then JAX.

TPU v4: optical circuit switches, SparseCores, and 4,096 chips

Mirrors route light between 64-chip cubes, so the pod's topology is chosen per job and routed around failures.

a TPU v4 pod: 64 cubes of 4 × 4 × 4 chips, joined by 48 optical circuit switches†



within a cube: electrical 3D mesh; between cubes: light, re-patched in milliseconds into a 3D torus (optionally twisted); the OCS layer is under 5% of system cost and under 3% of power (Jouppi et al., ISCA 2023)

275 TF

BF16 per chip, 7 nm

1.1 EF

4,096 chips, 3D torus

Three ideas in one machine†

Optical circuit switching: a job gets any slice shape (64 to 4,096 chips) and failed hosts are routed around, so slices stay usable at 99% host uptime.

SparseCore: dataflow engines for embedding lookups, about 5% of area and power, 5–7× on recommendation models; ads shaped the TPU as much as language did.

On-chip memory first: TPUv4i (ISCA 2021) put 128 MiB of CMEM on the inference die for performance per TCO.

Numbers: 275 TFLOP/s BF16, 32 GiB HBM2 at 1.2 TB/s, 192 W, 1.2–1.7× an A100 on Google's workloads; deployed 2020, Cloud 2022, paper 2023. XLA compiles one program for v2 through v4: the core model stayed while the pods changed.

v5e to TPU 8: an inference line, a training line, FP8 and FP4

Trillium quadrupled v5e; Ironwood added FP8 and 9,216-chip pods; generation 8 splits training from inference.

TPU†	v5e	v5p	v6e Trillium	v7 Ironwood	8t Sunfish	8i Zebrafish
announced	Aug 2023	Dec 2023	May 2024	Apr 2025	Apr 2026	Apr 2026
TensorCores	1	2	1	2	1	2
BF16 (TF)	197	459	918	2,307	n.p.	n.p.
FP8 (TF)	—	—	—	4,614	n.p.	n.p.
FP4 (PF)	—	—	—	—	12.6	10.1
HBM (GB)	16	96	32	192	216	288
HBM (TB/s)	0.8	2.8	1.6	7.4	6.5	8.6
ICI per chip	0.4 TB/s	1.2	0.8	1.2	2.4	n.p.
pod (chips)	256	8,960	256	9,216	9,600	n.p.
topology	2D torus	3D, OCS	2D torus	3D, OCS	3D torus	Boardfly

Google Cloud documentation and the Hot Chips 2025 Ironwood talk; TPU 8 figures from the April 2026 Cloud Next coverage (press); ICI as bidirectional bytes per second, the convention of the current docs (older blogs quote bits per direction); HBM usable capacity is slightly less (v5p 95 of 96 GiB); n.p.: not published

Reading the generations†

The fork (2023): “e” parts for cost and inference (one TensorCore, 256-chip 2D torus), “p” parts for training (two cores, SparseCores, OCS, 8,960 chips); Multislice joins pods over the data-center network.

Trillium (2024): 4.7× v5e per chip, double the HBM, still 256-chip pods; 99% weak scaling on GPT-3 175B in MLPerf Training 4.1 (Google’s figure).

Ironwood (2025): the first two-die TPU; native FP8 at 4.6 PFLOP/s, 192 GB HBM3e at 7.4 TB/s, 9,216 chips sharing 1.77 PB of HBM at 42.5 EFLOP/s; GA November 2025.

TPU 8 (2026): the parts diverge again: 8t keeps the torus and doubles ICI; 8i adds 384 MB of SRAM and a low-diameter “Boardfly” network for MoE all-to-all; 2× Ironwood’s perf/W (claim).

The TPU as a business: Google's own, then everyone's

For eight years the TPU was Google's own; in 2025–26 it became the first ASIC sold to others at gigawatt scale.



The stack that made it portable†

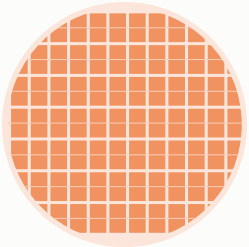
XLA (2017) compiles whole graphs to the VLIW cores; **JAX** (2018) made XLA the default for research; **Pathways** runs one program across pods; PyTorch/XLA and, from December 2025, TorchTPU with Meta bring PyTorch; Pallas exposes kernels. Google skipped MLPerf Training 5.1 (November 2025) and listed no TPU result in 6.0.

Why it matters for the rest of this lecture†

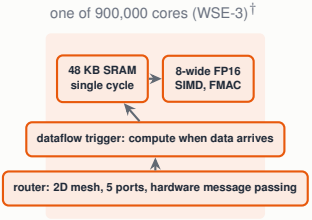
The TPU is the proof that a non-GPU accelerator can carry frontier training: Gemini trains on it, Anthropic and Meta buy it, Broadcom's XPU business is built on it. It also sets the terms: a systolic core, HBM, a torus with optical switching, a compiler that owns the graph, and a decade of patience. Every startup in the next sections had to answer why its approach would beat this, and NVIDIA.

Cerebras: the wafer is the chip

84 reticle fields wired across the scribe lines, spare cores for yield, 44 GB of SRAM on one 46,225 mm^2 die.



12 × 7 reticle fields, each a GPU-sized die



WSE†	1 (2019)	2 (2021)	3 (2024)
process	TSMC 16 nm	7 nm	5 nm
transistors	1.2 T	2.6 T	4 T
cores	400,000	850,000	900,000
SRAM, bandwidth	18 GB, 9 PB/s	40 GB, 20 PB/s	44 GB, 21 PB/s
dense FP16 (PF)	n.p.	7.5	about 15 (derived)
system	CS-1, 20 kW	CS-2, 23 kW	CS-3, 23 kW

Why wafer scale works, and what it costs†

Yield: a defect kills a core, not a wafer; about 1% spare cores and a fabric that maps bad ones out (Hot Chips 2019). Trilogy tried this in the 1980s and failed.

Why it helps: each core's weights sit in its own SRAM; 900,000 private SRAMs sum to 21 PB/s, 7,000× an H100's HBM, but not as shared bandwidth: the mesh is 27 PB/s and no core reads another's memory at SRAM speed.

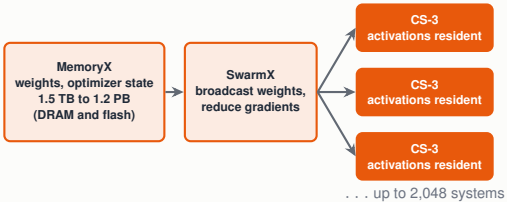
The core: small, in-order, dataflow-triggered, 8-wide FP16, skips zeros, 48 KB SRAM; 87M transistors/mm^2 against about 130 on Blackwell: most of the wafer is SRAM and wires.

Distrust "125 PFLOP/s": no precision or sparsity qualifier; dense FP16 is about 15 (derived from WSE-2's 7.5).

Weight streaming: MemoryX, SwarmX and Condor Galaxy

Activations stay on the wafer, weights stream in layer by layer from outside; clusters scale by data parallelism.

a Cerebras cluster for training†



each CS-3 holds a full copy of the activations and receives the same weights; only the batch is split, so one program fits any cluster size (Lie, IEEE Micro 2023); 24-trillion-parameter models claimed

Condor Galaxy†	CG-1	CG-2	CG-3
site, date	Santa Clara, Jul 2023	Stockton, Nov 2023	Dallas, 2024
systems	64 CS-2	64 CS-2	64 CS-3
“AI exaflops” (sparse)	4	4	8

What the design buys and gives up†

One program: no tensor, pipeline or expert parallelism to write; a GPT-3-class model in 565 lines of PyTorch (Cerebras’s figure); weights need not fit the wafer.

Data parallel only: every system computes every layer, so efficiency is linear in theory and the batch must grow with the cluster; gradients reduce through SwarmX.

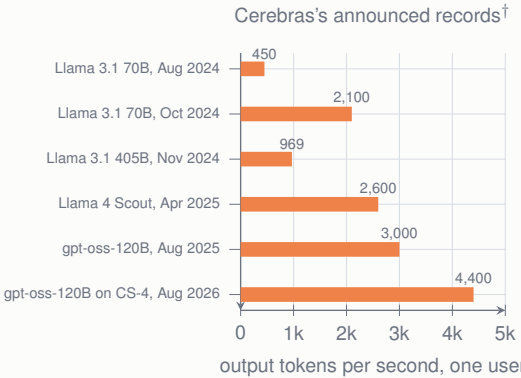
Customers: Argonne (first CS-1, 2019), LLNL, Sandia, PSC, EPCC, Mayo Clinic, GSK, G42/MBZUAI (Condor Galaxy); labs took the first units of each generation.

The business catch: G42, then MBZUAI, was 86–87% of revenue in 2024 and 2025 (S-1 filings), which delayed the first IPO attempt (slide 15).

No MLPerf: Cerebras has never submitted; its training claims are its own.

Cerebras Inference, the CS-4, and the IPO

From August 2024 the wafer became an inference engine; speed per user, not training, took it public.



vendor announcements citing Artificial Analysis; 16-bit weights at launch, lower precision later; GPU clouds served Llama 3.1 70B at 30–100 tok/s in 2024; one user's speed, not throughput

Timeline†

Aug 2024: Cerebras Inference: Llama 3.1 70B at 450 tok/s, "20× GPU clouds"; Mistral, Perplexity, Meta's Llama API and OpenAI's open models followed.

Sep 2024: S-1: 2023 revenue $78.7M, 83% from G42; CFIUS review; withdrawn October 2025 after a $1.1B round at $8.1B.

Jan–Apr 2026: OpenAI contract, over $20B and 750 MW through 2028, option to 2 GW; new S-1: 2025 revenue $510M, 86% still G42 and MBZUAI; backlog $24.6B.

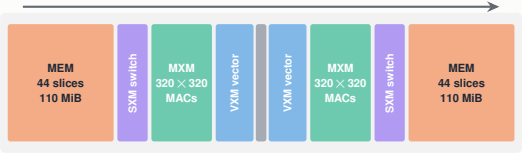
May 14, 2026: Nasdaq IPO (CBRS): $6.4B at $185; first close $311 ($95B diluted); ∼$96 by November.

Aug 2026: CS-4 "Nexus": three WSE-3T wafers (WSE-3 at twice the power) per rack, 750 "AI PFLOP/s"; CS-5 promised for 2027.

Groq: a deterministic tensor streaming processor

One core, 220 MiB of SRAM, no DRAM, every cycle scheduled by the compiler; Llama 2 70B took 576 chips.

the TSP die: functional slices span the chip, data streams east and west (Abts et al., ISCA 2020)†



320-wide vectors on 32 eastward and 32 westward streams; 144 instruction queues issue in lock step

Groq†	GroqChip 1 (TSP, 2020)	Groq 3 LPU (NVIDIA, 2026)
process, die	14 nm; 725 mm^2, 26.8B	Samsung 4 nm
SRAM	220 MiB; 55 TiB/s	about 500 MB; 150 TB/s
off-chip memory	none	none
peak, dense	750 TOPS INT8; 188 TF FP16	1.2 PF FP8
system	node 8, rack 72; to 10,440	LPX rack: 256 LPUs, 315 PF FP8

paper: 820 TOPS at 1 GHz, 900 MHz shipped; marketing quotes 80 TB/s of SRAM; FP16 uses two byte-planes, a quarter of the INT8 rate; LPX: 128 GB SRAM, 256 × 150 TB/s ≈ 40 PB/s (derived); no MLPerf result from Groq

From TSP to LPU to NVIDIA's LPX†

Design for determinism: no caches, arbiters or back-pressure; the compiler knows every cycle and is the scheduler. ISCA 2022 extended it to the network: scheduled links, error correction not retry, 722 ns per hop.

The LPU moment (Feb 2024): Llama 2 70B at 241 tok/s on Artificial Analysis, twice the next provider, from 576 chips in eight racks; Saudi Arabia committed $1.5B (2025).

No second chip: its Samsung 4 nm part (2023 deal) became NVIDIA's Groq 3; $750M at $6.9B (Sep 2025).

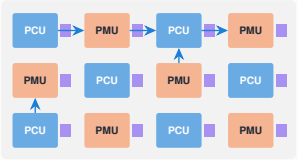
Dec 24, 2025: NVIDIA licensed the IP and hired Jonathan Ross and most engineers (reported $20B); Groq Inc. kept GroqCloud and re-raised at $3.5B (Aug 2026).

Groq 3 LPX (GTC 2026): decode FFN racks beside Vera Rubin, which does prefill, attention and KV; Gemma 4 31B at 3,431 tok/s (private endpoint); ships late 2026.

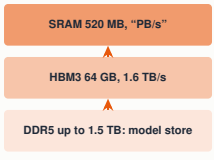
SambaNova: reconfigurable dataflow, three memory tiers

Compute and memory units wired per model by the compiler; SRAM, HBM and DDR hold a trillion parameters.

the RDU: a checkerboard of pattern compute units (PCU), pattern memory units (PMU) and switches†



a decoder layer becomes one fused pipeline; intermediates never leave the chip (SN40L: 1,040 PCUs and 1,040 PMUs on two dies)



SN40L's three tiers: experts and whole models swap from DDR into HBM in milliseconds

RDU†	SN10 (2020)	SN30 (2022)	SN40L (2023)	SN50 (2026)
process	7 nm; 40B	7 nm; 86B	5 nm; 102B	n.p.
SRAM	320 MB	640 MB	520 MB	432 MB
HBM	—	—	64 GB HBM3	64 GB HBM2E
DDR	1.5 TB DDR4	1 TB DDR4	1.5 TB DDR5	up to 2 TB
BF16 (TF)	325	688	638	about 1,600
system	SN10-8	SN30-8	SN40L-16, 10.2 PF	8 per chassis

Prabhakar et al., MICRO 2024 (arXiv 2405.07518); dense peaks; shipped SN40L-16 racks carry 768 GB DDR5 per RDU; SN50 is "5× SN40L" with FP8, about 1.6 PF BF16 (derived); no MLPerf result

Composition of experts, and the business†

The idea: a coarse-grained reconfigurable array (Stanford's Plasticine) onto which the SambaFlow compiler maps a whole dataflow graph, so operators fuse in space instead of running kernel by kernel through caches.

Samba-CoE (2024): 150 expert models, a trillion parameters on eight sockets, switched in milliseconds from the DDR tier; 3.7× a DGX H100 on that workload (paper). SambaNova Cloud (2024): Llama 3.1 405B at 132 tok/s in 16-bit on 16 RDUs.

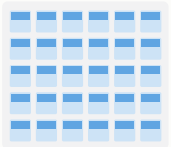
Customers: Argonne's ALCF AI Testbed (SN10, SN30), LLNL, RIKEN, Saudi Aramco, SoftBank, JPMorgan Chase.

Business: $676M at $5.1B (2021, SoftBank); a pivot to inference (2025); Intel's acquisition talks (Dec 2025) became a partnership and a $35M stake; $1B at $11B (Jul 2026) for SN50, shipping H2 2026.

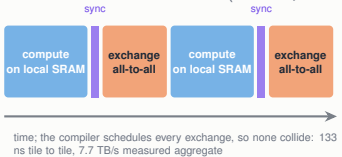
Graphcore: the IPU and bulk-synchronous parallelism

1,472 MIMD tiles with 900 MB of SRAM exchange data in compiler-scheduled supersteps; sold to SoftBank in 2024.

bulk-synchronous parallel execution on an IPU (Valiant, 1990; Jia et al., 2019)†



tiles: one core, six threads, 624 KB of SRAM each (GC200)



IPU†	GC2 (2018)	GC200 (2020)	Bow (2022)
process	16 nm	7 nm; 59.4B	7 nm + power die
tiles	1,216	1,472	1,472
SRAM	304 MiB	900 MB	900 MB
clock	1.6 GHz	1.325 GHz	1.85 GHz
FP16 dense (TF)	125	250	350
DRAM	host only	DDR4 (streaming)	same
systems	C2 card	M2000 to POD256	Pod16 to Pod1024

GC200 die 823 mm^2; the 45 TB/s SRAM figure is nominal (Citadel measured 30.7); IPUs publish no sparse peak, so compare with GPU dense FP16

What it proved, and what it did not†

The model: no shared memory, no caches, no DRAM on the first chip; independent tiles synchronize in bulk and exchange through a compiler-scheduled all-to-all. Flexible on sparse and irregular work; capacity bound by SRAM.

Bow (2022): the first production wafer-on-wafer part: a passive power-delivery die with deep-trench capacitors under the same logic lifted the clock 40% with no redesign.

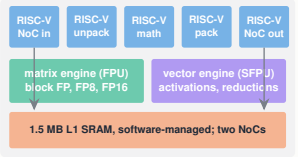
MLPerf Training v1.0–2.0: IPU-POD16 ResNet-50 in 28.3 min against 29.1 for a DGX A100 (v1.1); a POD16 listed at $149,995: the pitch was price per result, not peak.

The ending: the 10 EFLOP/s “Good computer” was never built; 2022 revenue $2.7M against a $205M loss; SoftBank bought the company in July 2024 for a reported $500–600M, below the 2020 valuation of $2.8B; no new silicon announced since.

Tenstorrent: Tensix cores, RISC-V and an open stack

Clusters of five small RISC-V cores around a matrix engine and 1.5 MB of SRAM, meshed; Ethernet on the die.

one Tensix “core” (Blackhole; Hot Chips 2024)†



$1,399

Blackhole p150, 32 GB

a mesh of 120–140 of these plus 16 “big” 64-bit RISC-V cores; no caches or shared memory

card†	Grayskull e150	Wormhole n300	Blackhole p150
cards shipped	2024	2024	2025
process	12 nm	12 nm, 2 dies	TSMC 6 nm
Tensix, clock	120 at 1.2 GHz	128 at 1.0	140* at 1.35
SRAM	120 MB	192 MB	180–210 MB
memory (GB/s)	8 GB LPDDR4, 118	24 GB GDDR6, 576	32 GB GDDR6, 512
FP8 / FP16 (TF)	332 / 83	466 / 131	745 / 372
Ethernet	none	16 × 100G per die	10–12 × 400G
power	200 W	300 W	300 W

datasheets, dense peaks; *140 Tensix designed, 120 enabled on shipping cards; Blackhole from Hot Chips 2024; the p150 page lists 664 TF “BLOCKFP8” at 120 cores

Not a core, not a GPU, not a closed stack†

A Tensix is a cluster: two RISC-V cores move data over the NoC, three issue to the shared matrix, vector and pack engines; “752 RISC-V cores” on Blackhole counts them, and 16 SiFive X280 cores let a board run without a host.

Ethernet-native scale-out: routers on the die join 32 chips into a Galaxy mesh (Wormhole 5.2 PF block FP8, 384 GB; Blackhole about 23.8 PF FP8, 1 TB GDDR6); no switch, no HBM, so the card stays cheap.

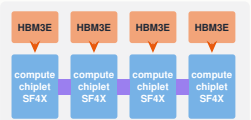
Open from kernels to compiler: TT-Metalium, TT-NN, TT-Forge (MLIR); the Ascalon RISC-V CPU is licensed as IP (LG, BOS; Rapidus 2 nm).

Money: $693M at about $2.6B (Dec 2024; Samsung Securities, Bezos, Fidelity, LG, Hyundai); $1.5B at $6.2B (Oct 2026); Galaxy “shipping commercially”; no MLPerf result.

Korea’s accelerators: Rebellions and FuriosaAI

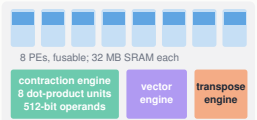
Latency-first inference chips on Samsung and TSMC 5 nm, then chiplets and HBM; the national fund’s picks of 2026.

REBEL-Quad: 4 chiplets, 4 HBM3E†



UCIe-Advanced die to die, about 4 TB/s aggregate; 144 GB HBM3E; one 2 PF FP8 device

RNGD: 8 processing elements of 64 slices†



one slice: tensor contraction, not matrix multiply, is the primitive (ISCA 2024)

chip†	Rebellions ATOM (2023)	REBEL-Quad (2025–26)	Furiosa RNGD (2024–26)
process	Samsung 5 nm	SF4X, 4 chiplets	TSMC 5 nm
peak, dense (TF)	32 FP16; 128 INT8	1,024 FP16; 2,048 FP8	512 FP8; 256 BF16
SRAM	64 MB	n.p.	256 MB
memory (GB; GB/s)	16 GDDR6; 256	144 HBM3E; 4,800	48 HBM3; 1,500
power (W)	60–150	350, then 600	150, then 180
MLPerf	Inference v3.0	—	none official

vendor briefs and Hot Chips 2025 (REBEL) and 2024 (RNGD); both chips’ power rose between brief and silicon; Furiosa’s Warboy (2021, 64 TOPS) ran MLPerf Inference v1.1; the entries are single-stream latency

Two paths out of the same market†

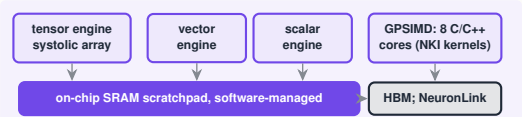
Rebellions (founded 2020): ATOM, a latency-critical SoC on GDDR6 (ISSCC 2024) for KT Cloud and SK Telecom; merged with SKT’s Sapeon (Dec 2024). REBEL answers the memory wall by packaging: four homogeneous chiplets on UCIe, each with an HBM3E stack (Hot Chips 2025). $250M at $1.4B (Sep 2025, Arm, Samsung); $400M at $2.3B (Mar 2026), the Korea National Growth Fund’s first pick; KOSPI IPO planned for 2027.

FuriosaAI (founded 2017): RNGD, a 180 W air-cooled card with two HBM3 stacks, adopted by LG AI Research for EXAONE (Jul 2025, 2.25× perf/W vs A100-class, LG’s figure); declined Meta’s reported $800M offer (2025); mass production Jan 2026; $620M at $2.9B (Oct 30, 2026); third generation with Broadcom. The CEO’s verdict: it needed four HBM stacks.

AWS Inferentia and Trainium: captive volume, four generations

A systolic NeuronCore, HBM and a torus; UltraServers reach 144 chips, and Anthropic's Rainier runs 500,000.

one NeuronCore (v2 onward): four engines and a scratchpad, no cache†



Trainium2: 8 NeuronCores on two dies with four HBM3 stacks; 16 chips per Trn2 instance in a 2D torus; Trainium3 adds a switched fabric

AWS chip†	Trainium	Inferentia2	Trainium2	Trainium3
GA	Oct 2022	Apr 2023	Dec 2024	Dec 2025
NeuronCores	2 (v2)	2 (v2)	8 (v3)	8 + comm.
dense peak	190 TF BF16	190 TF BF16	1.3 PF FP8	2.5 PF MXFP8
memory	32 GB HBM2e	32 GB HBM	96 GB HBM3	144 GB HBM3e
bandwidth	0.82 TB/s	0.82 TB/s	2.9 TB/s	4.9 TB/s
chip link	NeuronLink	384 GB/s	1 TB/s torus	2.6 TB/s
largest server	16 chips	12	64 (UltraServer)	144 (UltraServer)

AWS pages and blogs; Trainium2's 5.2 PF is with 4:1 sparsity; Trainium3 per chip derived from 362 PF per 144 chips; die sizes and TDPs not published; no MLPerf submissions

Why AWS builds its own†

Price-performance, not peak: AWS claims 30–40% better price-performance for Trn2 than H200 instances; the chip is sold only as EC2 capacity, so margin and supply are the product.

Scale-up grew 9× in a year: Trn2 instance 16 chips (20.8 PF FP8 dense), Trn2 UltraServer 64 (83 PF, 6 TiB HBM), Trn3 UltraServer 144 (362 PF MXFP8, 20.7 TB HBM3e, 706 TB/s).

Project Rainier (Indiana, $11B): nearly 500,000 Trainium2 live for Anthropic (Oct 29, 2025), "over one million by year end" as a target; Claude trains and serves on them.

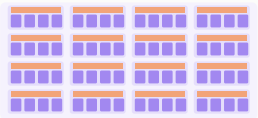
Trainium4 (Dec 2025): adopts NVLink Fusion, the first hyperscaler ASIC on NVIDIA's scale-up fabric.

Software: Neuron SDK (PyTorch, JAX); NKI for kernels.

Microsoft Maia and Meta MTIA: inference first, then everything

Maia runs on Ethernet; MTIA grew from a 25 W recommendation chip into a 1.7 kW HBM part in three years.

Maia 100 (Hot Chips 2024): 16 clusters of 4 tiles on one reticle-sized die; Ethernet out of every chip†



tile: tensor unit (MX formats) + vector cluster SRAM

chip†	Maia 100	Maia 200	MTIA v1	MTIA v2	MTIA 400
announced	Nov 2023	Jan 2026	May 2023	Apr 2024	Mar 2026
process	N5	N3P	7 nm	5 nm	chiplets
die (mm²)	820	n.p.	373	421	—
dense peak	0.8 PF BF16	5.1 PF FP8	102 TOPS	354 TOPS	6 PF FP8
on-chip SRAM	500 MB	272 MB	128 MB	256 MB	n.p.
memory	64 GB HBM2e	216 GB HBM3e	64 GB LP5	128 GB LP5	288 GB HBM
bandwidth	1.8 TB/s	7 TB/s	176 GB/s	205 GB/s	9.2 TB/s
power	700 W	750 W	25 W	90 W	1,200 W
fabric	Eth 4.8 Tb/s	Eth 2.8 TB/s	PCIe 4	PCIe 5	72/rack

Microsoft blogs and Hot Chips 2024 (Maia 100 also quoted at 3 POPS for 6-bit MX); MTIA from ISCA 2023 and Meta's blogs via press; LP5: LPDDR5

Two companies, two reasons†

Microsoft: tokens per dollar. Maia 100 (2024): a reticle-limit N5 die, MX block formats, an Ethernet scale-up fabric; little use beyond internal tests. Maia 200 (Jan 2026: N3P, 216 GB HBM3e, 10 PF FP4) uses standard FP4/FP8, serves GPT-5.2 in 6,144-chip domains, and runs only inside Azure; it slipped a year (press).

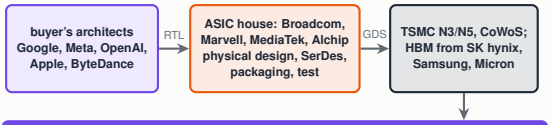
Meta: ranking first, now everything. MTIA v1 (ISCA 2023): an 8 × 8 grid of dual-RISC-V elements, 128 MB SRAM, LPDDR5, 25 W, for recommendation inference; v2 kept the grid at 3.5× the throughput. The 2026 family (MTIA 300–500, with Broadcom): HBM chiplets, 72-chip switched racks, 1.2–1.7 kW, aimed at generative AI; Meta also rents TPUs and buys AMD.

In common: PyTorch and Triton back ends, no MLPerf results, the owner's own workloads.

OpenAI–Broadcom and the custom-XPU model

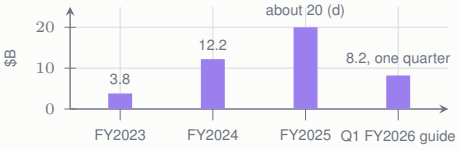
The buyer owns the architecture, Broadcom the silicon and SerDes, TSMC the wafers; 10 GW for OpenAI by 2029.

how a custom accelerator gets built (2023–26)†



the buyer keeps the architecture and the software; the ASIC house turns it into a chip in 12–18 months and sells it back at a margin far below NVIDIA's

Broadcom AI semiconductor revenue (XPUs and Ethernet)†



The deals, and what they buy†

OpenAI–Broadcom (Oct 13, 2025): 10 GW of custom accelerators and racks, OpenAI designing chip and system, Broadcom building and deploying over Ethernet from H2 2026 to 2029; Reuters: a TSMC 3 nm systolic design with HBM, a team of about 40 led by a TPU veteran. No shipped rack reported by Nov 2026; OpenAI also signed 10 GW with NVIDIA and 6 GW with AMD.

Broadcom's book: five XPU customers: Google's TPU since v1, Meta's MTIA, and a $10B order plus $11B for late 2026 from one customer, reported to be Anthropic's TPUs; Apple and ByteDance are reported. Marvell does the same for AWS and Microsoft.

What it buys: the owner's workload baked in, HBM and fabric sized to it, a part sold near cost. **What it costs:** two to three years, a compiler team, all the risk.

The established vendors: Gaudi, Cloud AI, Spyre, Dojo

Intel bought one line and killed another; Qualcomm bets on LPDDR; IBM sits beside the mainframe; Tesla paused.

chip†	Gaudi 2	Gaudi 3	Cloud AI 100 Ultra	IBM Spyre	Tesla D1
vendor, year	Intel 2022	Intel 2024	Qualcomm 2023	IBM 2025	Tesla 2021
process	7 nm	5 nm, 2 dies	7 nm, 4 SoCs	5 nm	7 nm
units	24 TPC, MME	64 TPC, 8 MME	64 AI cores	32 cores	354 nodes
dense peak	432 TF BF16	1,835 TF	870 TOPS	300 TOPS	362 TF BF16
on-chip SRAM	48 MB	96 MB	576 MB	64 MB	440 MB
memory	96 GB HBM2e	128 GB HBM2e	128 GB LP4x	128 GB LP5	none on die
bandwidth	2.46 TB/s	3.67 TB/s	548 GB/s	n.p.	2 TB/s edge
power	600 W	900/1,200 W	150 W	75 W	400 W
scale-out	24 × 100 GbE	24 × 200 GbE	PCIe card	8 per drawer	25-die tile

Intel whitepapers and Hot Chips 2024 (Gaudi 3 gives one figure for BF16 and FP8); Qualcomm's Ultra brief; IBM at Hot Chips 2024 (Samsung 5 nm; no LPDDR bandwidth); D1 from Hot Chips 34 (TSMC; 2 TB/s per edge, 2021 slides said 4), TDP from the 2021 AI Day via press; LP: LPDDR

$2B

Habana, Dec 2019

200 MW

Humain on AI200, 2026

Four fates†

Intel Gaudi: Nervana (bought 2016) was cancelled in Jan 2020 for Habana (about $2B, Dec 2019). Gaudi 2 was the only merchant non-NVIDIA part with GPT-3 results in MLPerf Training (2023–24); Gaudi 3 never appeared and missed a $500M 2024 target.

Qualcomm Cloud AI: a scaled-up phone NPU with big SRAM and LPDDR, a steady MLPerf Inference submitter 2021–24; AI200 (2026) puts 768 GB of LPDDR per card in 160 kW racks, AI250 (2027) adds near-memory compute.

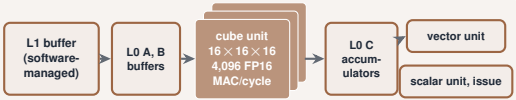
IBM Spyre: 32 cores at 75 W beside Telum II in z17 and Power11 (GA late 2025), inference next to the transactions.

Tesla Dojo: a DRAM-less mesh of 354 nodes per die, 25 dies per tile; shipped 2023, cut Aug 2025, restarted Jan 2026.

Huawei Ascend: Da Vinci cores to the 950/960/970 roadmap

Cube-MAC cores on 7 nm; cut off from TSMC in 2020, Huawei designs its own HBM and scales by count.

one Da Vinci AI core (Hot Chips 31, 2019): a 3-D cube of multipliers with vector and scalar side units†



30–32 cores per 910 die (456 mm², TSMC 7 nm); 20–25 active on the SMIC-built 910B; two 910B dies make a 910C

Ascend†	910	910B	910C	950PR	950DT	960
ships	Aug 2019	2023	2025	Q1 2026	Q4 2026	Q1 2027
process	TSMC 7 nm	SMIC N+2	2 dies	n.p.	same die	n.p.
FP16 (TF)	256–320	up to 400	about 800	—	—	—
FP8 (PF)	—	—	none	1	1	2
FP4 (PF)	—	—	—	2	2	8 (PR)
HBM	HBM2	64 GB, 1.6 TB/s	128 GB	HiBL 1.0	144 GB, 4 TB/s	about 2×
link (TB/s)	n.p.	n.p.	0.8	2	2	2.2

910: 256 TFLOP/s at launch, 320 on the top SKU (CSET); 910B/C are analyst estimates, no datasheet; 950/960 from Huawei Connect 2025 and 2026 (own HBM: HiBL 1.0, HiZQ 2.0); 910C link derived from "2.5×" for the 950PR; Ascend 970 (2028): 3.6 PF FP8, 14 PF FP4

From launch to the roadmap†

Da Vinci (2019): a TPU-like cube core replicated across the die; CANN is the CUDA-equivalent layer (open-sourced Aug 2025); one MLPerf entry, 512 chips in Training v0.7 (2020).

Cut off: the Entity List (2019) and the foreign direct product rule (2020) ended TSMC supply; the 910B moved to SMIC's 7 nm-class N+2 with fewer cores (SemiAnalysis: most 910B/C dies still came from TSMC via Sophgo, with stockpiled Samsung HBM).

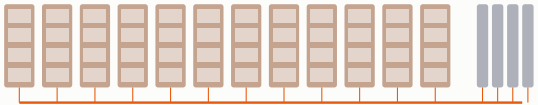
The DeepSeek test: R2 training on Ascend never completed a run (FT, Aug 2025); training went back to H20s.

The roadmap (Xu, 2025): the mainland will "lag in process nodes for a long time"; so own HBM, FP8/FP4, fabric scale (slide 26).

CloudMatrix 384 and the SuperPoDs: scale, not per-chip speed

384 Ascend 910C on an all-optical fabric out-total a GB200 NVL72 at about four times the power.

CloudMatrix 384 (Atlas 900 A3 SuperPoD, 2025): 16 racks, 6,912 optical transceivers†



12 compute racks × 32 Ascend 910C = 384 NPUs (left), 4 switch racks (right); all-to-all optical scale-up on 400G linear-drive transceivers, UnifiedBus 1.0

2025 racks†	CloudMatrix 384	GB200 NVL72	ratio
accelerators	384 Ascend 910C	72 B200	5.3×
dense BF16 (PF)	about 300	180	1.7×
HBM (TB)	49	13.4	3.6×
HBM bandwidth (TB/s)	about 1,230	576	2.1×
scale-up fabric	optical, all-to-all	NVLink 5, copper	—
power, all-in	∼559 kW	∼145 kW	3.9×

SemiAnalysis (April 2025) for the CloudMatrix figures and ratios (per FLOP 2.3× the power, per TB/s 1.8×, per TB 1.1×); NVL72 from NVIDIA's page (BF16 dense = 72 × 2.5 PF); 910C values inferred from totals

Reading the comparison, and what comes next†

Aggregate, not per chip: 5.3× the accelerators give 1.7× the dense BF16; a 910C is about a third of a B200 and has no FP8. Memory is the real win: 3.6× the HBM for MoE weights and KV cache.

The exchange rate: about 3.9× the power, 6,912 transceivers to keep alive, 16 racks of floor: NVIDIA's NVL72 move, at a worse price.

Deployed: deliveries from March 2025; over 300 systems by September 2025 (Huawei). **UnifiedBus 2.0** (2025): 200 m optical reach, 100 ns failover, 2.1 μs between cabinets.

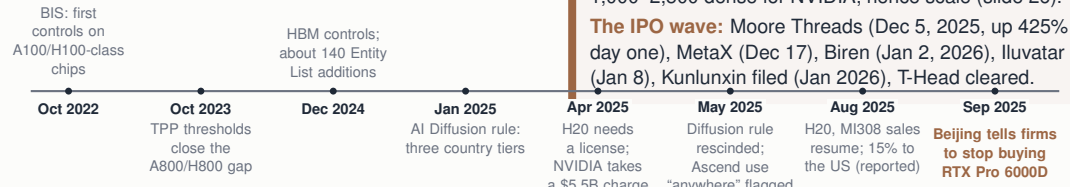
SuperPoDs: Atlas 950 (Q4 2026): 8,192 Ascend 950DT, 8 EF FP8, 1,152 TB, 16 PB/s; Atlas 960 (2027): 15,488 chips, 30 EF FP8; SuperClusters to a million NPUs; announced, not shipped.

Kunlun, Hanguang, Cambricon and the policy backdrop

China's ASIC lines, its GPU-like rivals' IPO wave, and the export-control calendar behind both.

chip†	year	peak	note
Baidu Kunlun 2	2021	128 TF FP16	7 nm, about 120 W
Kunlun P800	2025	∼345 TF FP16	30,000-card cluster (Apr 2025)
Kunlun M100, M300	2026–27	n.p.	inference; training; 512-card node
Alibaba Hanguang 800	2019	820 TOPS INT8	12 nm, 17B transistors; inference
Alibaba T-Head PPU	2025	"H20 class"	96 GB HBM2e; 16,384-card order
Cambricon MLU370	2021	96 TF FP16	last pre-sanction part
Cambricon MLU590	2023	∼345 TF FP16	adds FP8; 690 slipped to 2026

press and analyst figures (Caixin, Recode China AI, TrendForce); the GPU-like designs (Biren, Moore Threads, MetaX, Hygon, Iluvatar) are left out as GPUs



What the controls made†

Demand: each rule moved buyers to domestic parts: Cambricon went from a 2024 loss to RMB 2.06B profit on 2025 revenue up 453%; Baidu runs 30,000 P800s; Alibaba's PPU took a 16,384-card order.

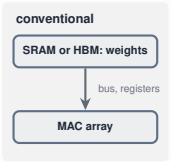
Ceiling: SMIC 7 nm by DUV multipatterning, low yield, HBM from stockpiles: 100–400 TF FP16 per chip against 1,000–2,500 dense for NVIDIA; hence scale (slide 26).

The IPO wave: Moore Threads (Dec 5, 2025, up 425% at day one), MetaX (Dec 17), Biren (Jan 2, 2026), Iluvatar (Jan 8), Kunlunxin filed (Jan 2026), T-Head cleared.

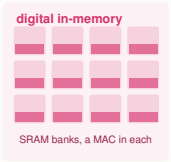
Compute in the memory: d-Matrix, Untether, Mythic, EnCharge

Put the multiply in the SRAM array and the bytes stop moving; capacity and precision are the price.

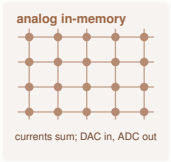
three places to put the multiply-accumulate



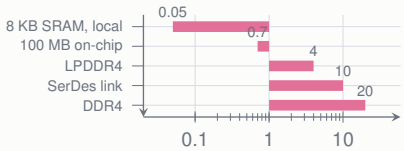
every weight travels across the chip, or from DRAM, on every use



weights stay put; exact integer math (d-Matrix, Untether, NorthPole)



Kirchhoff sums the dot product; drift and converters bound precision



energy per bit moved, pJ (Dally, Turakhia, Han, CACM 2020)

The parts, and where they stand†

d-Matrix Corsair (Nov 2024): eight 6 nm chiplets, 2 GB of SRAM in all, arrays repurposed as multipliers, 150 TB/s per card, 256 GB of LPDDR5 behind; “9.6 POPS” at 4-bit block floating point; 64 cards hold a 100B model in SRAM; Raptor (2026) stacks DRAM. $275M at $2B, Nov 2025.

Untether speedAI240: compute in the SRAM arrays, 1,458 RISC-V cores; a rare MLPerf power submitter; wound down, team to AMD, Jun 2025.

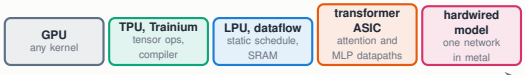
Analog: Mythic M1076 keeps weights as flash conductances; out of cash Nov 2022; $125M raised Dec 2025. EnCharge EN100 (2025): charge-domain, for clients; no independent data. Qualcomm AI250 (2027): “near-memory”, no detail (slide 24).

The trade: digital IMC keeps exact math but caps capacity at gigabytes; analog saves energy, loses precision.

The specialization spectrum: from GPU to hardwired model

Each step right trades freedom for speed; Taalas etched a model into metal and was bought in six months.

what is fixed at design time, left to right



chip†	what it is	the claim	Nov 2026
Etched Sohu	transformer-only ASIC, TSMC N4P, 144 GB HBM3E	500k tok/s, Llama 70B, 8 chips	A0 silicon Jun 2026, no benchmark
MatX One	split systolic array; SRAM weights, HBM KV cache	2,000 tok/s on a large MoE	no silicon; $500M Feb 2026
Positron Atlas	Agilex 7 FPGA + HBM + DDR5	price per token	50+ racks at Oracle
Positron Asimov	ASIC, N3P, up to 2.3 TB LPDDR5X, 3 TB/s	capacity first	tape-out end 2026, volume 2027
Taalas HC1	Llama 3.1 8B in metal: 6 nm, 815 mm^2, 3/6-bit	17,000 tok/s, one user	demo Feb 2026; AMD deal, Aug 2026

vendor figures, one user except Etched (server); no MLPerf; Asimov simulated

Claims against shipments†

Money before silicon: Etched raised $1B at $10–21B (Jul–Aug 2026) with no independent number; MatX $500M (Feb 2026), no chip shown; Positron $875M at $5B (Sep 2026) before Asimov taped out. Only the FPGA Atlas ships.

What the transformer ASIC removes: general matrix flexibility, for attention- and MLP-shaped datapaths; the bet is that the architecture stops changing (MoE and new attention variants are the risk).

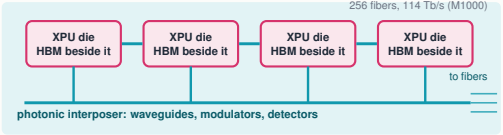
The endpoint: Taalas puts weights in the metal layers: a new model is a new mask set, precision is 3–6 bits, one user sees 17,000 tokens per second; AMD agreed to buy it in Aug 2026.

Exits: NVIDIA licensed Groq (slide 16); AMD took Untether's team and Taalas; Marvell, Celestial (slide 30).

Light and spikes: photonics and neuromorphic computing

Photonic compute was deferred and money went to interconnect; neuromorphic machines stay research tools.

a photonic interposer: electrical on top, light between packages (Passage M1000, Photonic Fabric)†



project†	what and how much	status, Nov 2026
Lightmatter Envise	photonic matrix-vector multiply	deferred to 2030+
Lightmatter Passage	M1000 interposer, 114 Tb/s, 4,000 mm^2; L200 CPO, up to 64 Tb/s per chip	NPO 2026, CPO 2027
Celestial AI	Photonic Fabric chiplet, 16 Tb/s, stacks on hot dies	Marvell buys it, Dec 2025 ($3.25B+)
Ayar Labs	TeraPHY optical I/O chiplets	valued above $1B, Dec 2024
Intel Hala Point	1,152 Loihi 2, 1.15B neurons, 20 POPS, 2.6 kW	Sandia research system, Apr 2024
IBM NorthPole	12 nm, 22B transistors, 256 cores, 224 MB, no DRAM	prototype; 3B LLM on 16 cards

Why these stay niche, and what to watch†

Photonic compute multiplies in the analog domain; precision, conversion energy and no optical memory shelved Envise. **Interconnect** is where light wins: a SerDes costs about 10 pJ per bit (Dally, 2020), so rack traffic moves to fiber at the package: NVIDIA's 2026 switches use co-packaged optics, Marvell paid up to $5.5B for Celestial.

Neuromorphic energy scales with spikes, not multiply-adds; Hala Point's 20 POPS hold for sparse work and no spiking machine has shown dense LLM throughput.

NorthPole is the near-memory limit case: 224 MB and no DRAM run a 3B model at under 1 ms per token, 72.7× an H100's tokens per joule at low latency (IBM's own measurement).

The chips side by side: what each vendor published

Dense peaks, memory and fabric for eleven non-GPU parts against one GPU; the gaps (n.p.) are part of the picture.

chip† / vendor	B200 NVIDIA	TPU v7 Google	Trn2 AWS	Maia 200 Microsoft	MTIA v2 Meta	WSE-3 Cerebras	SN40L SambaNova	LPU Groq	Bow Graphcore	Blackhole Tenstorrent	RNGD Furiosa	910C Huawei
first shipped	2024	2025	2024	2026	2024	2024	2024	2020	2022	2025	2026	2025
process	TSMC 4NP	n.p.	n.p.	TSMC N3P	TSMC N5	TSMC N5	TSMC N5	GF 14 nm	N7, WoW	TSMC N6	TSMC N5	SMIC N+2
organization	148 SMs	2 TensorCores	8 cores	∼96 cores	64 PEs	900k cores	1,040 PCUs	1 core	1,472 tiles	140 Tensix	8 PEs	2 dies
BF16 dense (TF)	2,250	2,307	667	1,270	177	∼15,000	638	188	350	372	256	∼800
FP8 dense (TF)	4,500	4,614	1,300	5,070	—	—	—	—	—	745	512	—
FP4 dense (TF)	9,000	—	—	10,150	—	—	—	—	—	—	—	—
INT8 dense (TOPS)	4,500	—	—	—	354	—	—	750	—	—	—	—
on-chip SRAM	n.p.	n.p.	n.p.	272 MB	256 MB	44 GB	520 MB	230 MB	900 MB	210 MB	256 MB	192 MB
memory	HBM3e	HBM3e	HBM3	HBM3e	LPDDR5	none	HBM3, DDR5	none	none	GDDR6	HBM3	HBM2e
capacity (GB)	180	192	96	216	128	—	64, 1,536	—	—	32	48	128
bandwidth (TB/s)	7.7	7.4	2.9	7.0	0.2	—	1.6	—	—	0.5	1.5	3.2
scale-up (GB/s)	1,800	1,200	1,000	2,800	PCIe	150	n.p.	480	64/link	400	128	n.p.
largest domain	8	9,216	64	6,144	72	2,048	16	10,440	1,024	32	8	384
power (W)	1,000	n.p.	n.p.	750	90	23,000	n.p.	∼300	n.p.	300	180	n.p.
software	CUDA	XLA, JAX	Neuron	Maia SDK	PyTorch	CSoft	SambaFlow	GroqWare	Poplar	TT-Metal	Furiosa	CANN

vendor figures (datasheets, Hot Chips talks, papers), dense unless marked; B200 is the HGX B200 module; MTIA v2 FP8 is not published (INT8 shown); WSE-3 BF16 is derived from the WSE-2's 7.5 PF (Cerebras publishes only "125 PF AI compute"), its power is the CS-3 system and its 150 GB/s is the CS-3 host I/O; SN40L memory is 64 GB HBM3 plus up to 1.5 TB DDR5; LPU power from third-party listings; Bow has no on-package DRAM (weights stream from host DDR); Blackhole figures at 140 cores (shipping SKUs run 120); 910C memory and bandwidth are analyst estimates (two 910B dies); scale-up is the per-chip link into the largest coherent domain in the next row; n.p.: not published as of November 2026.

MLPerf: who shows up, and who never did

The neutral chip benchmark: TPUs, Gaudi, the IPU and a few inference parts came; Cerebras and Groq never did.

non-GPU submissions to MLPerf by year (closed and open divisions; filled = Training, hollow = Inference)†



notable result†	what was shown
Huawei, Training v0.7 (2020)	512 Ascend 910 on ResNet-50; four of five open-division results
Graphcore, Training v1.0 (2021)	IPU-POD64: BERT about 9 min, ResNet-50 14.5 min; pitched on price (POD16 $150k)
Qualcomm, Inference v3.0 (2023)	18 Cloud AI 100 Pro in 2U: 430k ResNet-50 inferences/s, 241 per watt
Intel, Training v4.0 (2024)	GPT-3 175B on 1,024 Gaudi 2 in 66.9 min; Gaudi 3 never appeared
Google, Training v4.1 (2024)	Trillium preview and v5p pods on GPT-3 175B, 99% weak scaling (Google's figure)

Reading the matrix†

Who submits: vendors selling a general part against an existing GPU number: Google (Training v0.5–5.0 bar v2.1, v3.0), Habana then Intel (2021–24), Graphcore (2021–22), Qualcomm (2021–24).

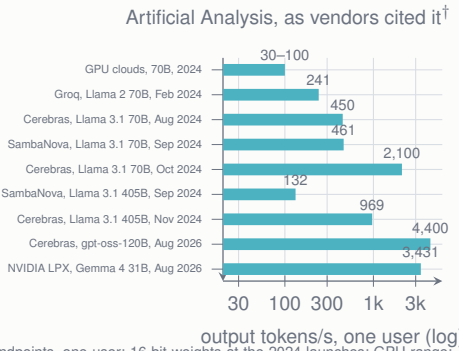
Who left: Graphcore after v2.0, Intel after Gaudi 2, Qualcomm after v4.0, Untether after one round; Google skipped Training 5.1 and has not entered Ironwood.

Who never came: Cerebras, Groq, SambaNova, Tenstorrent, the captive chips (Trainium, Maia, MTIA) and the Chinese parts after 2020. These three sell latency per user, which time-to-train and throughput metrics do not reward; captive chips have no buyer to convince.

What it means: most numbers in this lecture are the vendors' own or a third party's service measurements (slide 33); a peak without a benchmark is a claim.

Inference speed and price: what the SRAM machines deliver

Thousands of tokens per second for one user, third-party measured; the catch is the cost per token.



public endpoints, one user; 16-bit weights at the 2024 launches; GPU range: typical 2024 cloud serving at batch; LPX: a private pre-release endpoint at concurrency 1 (Artificial Analysis for NVIDIA); the CS-4 figure is the vendor's

Speed, price and the batch†

Why they are fast: the weights sit in SRAM, so one user's decode step never waits on HBM (slide 8); the limit moves to pipeline depth and chip-to-chip latency. Cerebras's 70B went 450 → 2,100 tok/s in two months, on software.

What it costs: capacity is bought in chips: Groq served Llama 2 70B from 576. SemiAnalysis (Feb 2024): Groq ahead on silicon cost per token at low latency, NVIDIA about 10× better per dollar at high throughput (the batch amortizes each weight read).

The prices: Mixtral 8x7B at $0.27 per million tokens on Groq (2024); Cerebras's 405B at $6 in, $12 out; gpt-oss-120B (Nov 2026) at $0.39/M on Cerebras against $0.04–0.05 on GPU clouds (first answer 2.1 s vs 6–69 s).

The caveat: a service is measured, not a chip: quantization, speculation and batching are the provider's.

Deals and share, 2025–26: commitments are not deployments

The non-GPU chips won gigawatt-scale orders for the first time; most of that power is not yet installed.

commitment†	what	status, Nov 2026
Anthropic–Google (Oct 2025, Apr 2026)	up to 1M TPUs, over 1 GW in 2026; Broadcom booked $21B of TPU orders; multiple GW more from 2027	Ironwood GA; racks delivering
Anthropic–AWS Rainier (2025)	$11B Indiana campus; about 500k Trainium2 live Oct 2025 (1M+ AWS-wide); 2.2 GW planned	live
OpenAI–Broadcom (Oct 2025)	10 GW of custom accelerators and racks, H2 2026–2029	no racks reported
OpenAI–Cerebras (Jan, Apr 2026)	over $20B, 750 MW through 2028, option to 2 GW	CS-3/CS-4 shipping
Meta–Google (Feb 2026)	multibillion TPU rental via Google Cloud; purchases discussed for 2027	rental
KSA, Humain	$1.5B Groq LPU expansion (Feb 2025); 200 MW of Qualcomm AI200/AI250 from 2026	Groq live; Qualcomm announced
Microsoft Maia 200 (Jan 2026)	inference in two Azure regions; no external sale	in production
Huawei CloudMatrix 384	over 300 systems delivered by Sep 2025 (about 115k 910C, derived)	shipping

How much of the market†

Share: TrendForce (Oct 2025): NVIDIA about 70% of the 2025 AI-chip market; custom-ASIC *unit* shipments to pass GPUs in 2026; by value GPUs dominate, since a TPU or Trainium sells only as cloud capacity.

Broadcom as the proxy: AI revenue $12.2B in FY2024 (up 220%), about $20B in FY2025 (derived), $8.2B guided for one FY2026 quarter; five XPU customers.

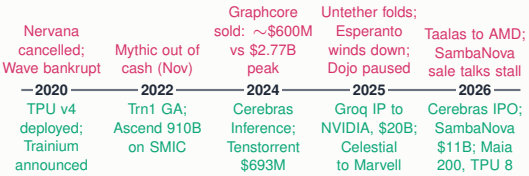
Startups: a record $6.2B of US semiconductor venture funding in 2025 (half Cerebras, Groq, PsiQuantum); then the exits: Groq's technology to NVIDIA for about $20B (Dec 2025), Cerebras's $6.4B IPO (May 2026), SambaNova at $11B, Tenstorrent at $6.2B (2026).

Read the table as power contracted, chips live, racks reported: only Rainier, Ironwood, Maia 200 and CloudMatrix are deployments; the 10 GW is a plan.

Why most of them failed, and what the survivors share

A few found captive volume, a niche or a buyer for their IP; the rest ran out of software or money.

exits, 2020–2026: above the line by failure, below it by success (dated events; sources on the section slides)†



The pattern: a chip 2–5× better on one axis arrives a year late against a GPU whose software already runs the new model; a buyer who owns the workload and the compiler (Google, Amazon, Meta, Huawei) can wait that year, a merchant startup cannot. Groq, Celestial, Taalas and Untether ended as engineering teams inside larger companies.

Five reasons, and three ways out†

1. Software: each design needed a compiler for a static schedule plus the libraries users expect; CUDA had a fifteen-year start (slide 6).

2. Memory: SRAM-only parts capped the model per chip; HBM parts queued behind NVIDIA for supply (slides 8, 31).

3. Scale-out: no NVLink or NCCL equivalent; startups sold nodes while training went to 10^4 chips (slides 10, 26).

4. The moving target: CNNs, transformers, MoE, long context; a fixed datapath fits the last one (slides 29, 32).

5. Capital: a tape-out every 18–24 months at nine figures (slide 34).

Ways out: *captive volume* (TPU, Trainium, MTIA, Maia, Ascend); *a niche* (speed per user: Cerebras, Groq); *becoming IP* (Groq, Tenstorrent, Celestial, Taalas).

Summary: what to take from this lecture

The rivals won where someone else owned the workload; the 2027 questions are about scale, not cores.

Eight things to remember

1. Decode is bandwidth-bound; every design here answers that (slide 4).
2. Five organizations: systolic, dataflow, streaming, MIMD mesh, GPU (slide 5).
3. The TPU is the proof: systolic cores, HBM, optical torus, a compiler that owns the graph (slides 9–12).
4. The wafer works, inference first (slides 13–15).
5. SRAM streaming buys latency, costs capacity (slides 16, 33).
6. Hyperscaler chips are the volume: 1M+ Trainium2, TPUs by the gigawatt (slides 21–23).
7. China scales out what it cannot scale up: 384-chip SuperPoDs, in-house HBM (slides 25–27).
8. Distrust “AI FLOPS”: which precision, dense or sparse, measured by whom (slides 31–33).

Open questions for 2027

Do the custom chips ship at the promised scale? The OpenAI–Broadcom 10 GW, Trainium 3 and 4, Maia 200 and TPU 8 are commitments; deployments will show whether ASICs pass GPUs in units (TrendForce).

Does inference speed stay a business? Cerebras and the NVIDIA LPX rack sell speed per user; GPUs with FP4 and more HBM close the gap each year.

Does Ascend reach its roadmap? 950DT in Q4 2026, 960 in 2027, 8,192-chip SuperPoDs, on SMIC 7 nm-class processes and in-house HBM.

Which startups are left? SambaNova’s SN50, Rebellions’ REBEL systems and d-Matrix’s Raptor ship or slip in 2027.

Does the GPU absorb the ideas? Tensor memory and an LPU in NVIDIA’s rack were the rivals’ ideas.

Sources

Papers and talks, vendor documentation, filings, benchmark tables and reporting; † marks vendor or press material.

Concepts and the TPU
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Cerebras, Groq, SambaNova†
Lie, “Cerebras architecture deep dive,” IEEE Micro 2023, and Hot Chips 2019–2026 talks; Cerebras S-1 filings (2024, 2026) and press releases (WSE-3, CS-4, OpenAI); Abts et al., “Think fast: a tensor streaming processor,” ISCA 2020, and “A software-defined tensor streaming multiprocessor,” ISCA 2022; Groq product briefs; Prabhakar et al., “SambaNova SN40L,” MICRO 2024; Peng et al. on SN30; SambaNova releases; Emani et al., PMBS 2022 and arXiv 2310.04607 (ALCF AI Testbed evaluations).

Graphcore, Tenstorrent, Rebellions, Furiosa†
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Hyperscalers†
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China†
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Emerging†
d-Matrix Corsair and Raptor materials; Untether speedAI; Mythic; EnCharge; Etched, MatX, Positron and Taalas announcements; Lightmatter Passage; Marvell–Celestial release; Davies et al., Loihi 2; Modha et al., NorthPole, Science 2023.

Measured and money
MLCommons Training v0.5–6.0 and Inference v0.5–6.0 result tables; Artificial Analysis provider benchmarks; SemiAnalysis, “Groq inference tokenomics” (2024); TrendForce (Oct 2025); Crunchbase News (Jan 2026); company filings and press releases for the deals table. Full list with links in the paper kit.